

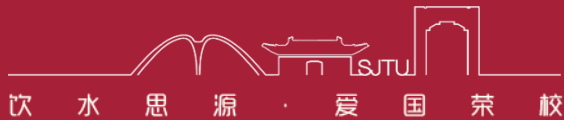


上海交通大学
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Prioritization of self-associated voices is enhanced by positive prosodic valence: Roles of individual explicit self-esteem in self-bias and positive self-bias

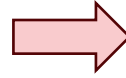
Presenter: Mou Yifan, Ding Hongwei

Mou, Y., & Ding, H. (2026). Prioritization of self-associated voices is enhanced by positive prosodic valence: Roles of individual explicit self-esteem in self-bias and positive self-bias. *Consciousness and Cognition*, 139, 103995.



□ Background

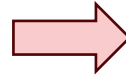
- limited cognitive capacity and attention
- challenges from the vast amount of information



- individuals **selectively allocate attention** to socially salient information (Sui & Humphreys, 2015).

Self-relevant information

- Cocktail party effect (Cherry, 1953)
- Dichotic listening (Moray, 1959)



- Not only restricted to self-names
- But also generalize to:
 - Self-faces
 - Own body parts
 - Self-generated actions
 - Self-owned objects
 - (Frassinetti et al., 2011; Golubickis et al., 2021; Sui & Humphreys, 2013)
- What about self-voices?



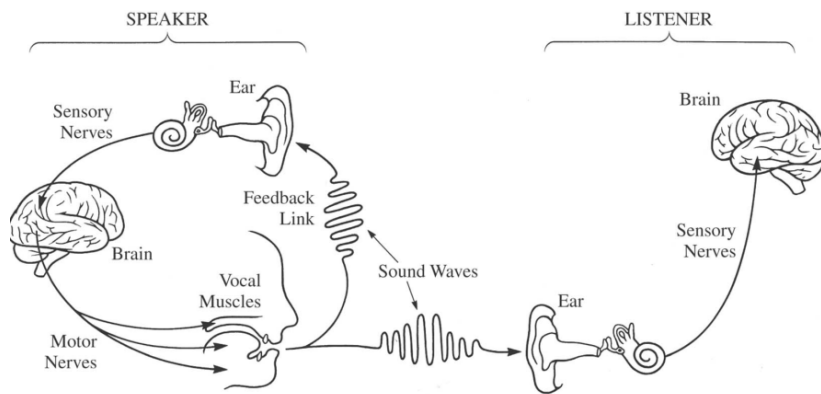
□ Background

Distinctive characteristics of Self-voice

- Unconsciously **passive exposure**
- Requirement of **self-generation**
- **Multimodal** transmissions
 - Air-conducted
 - Bone-conducted

Important characteristics of Self-voice

- Carrier of rich **linguistic information**
- Carrier of **non-linguistic information** (e.g., social identity)
- The most frequently encountered sound
- As the most frequently encountered sound daily, self-voice is central to the construction of self-identity and social interactions (Payne et al., 2024; Pinheiro et al., 2023).



□ Self-voice advantages

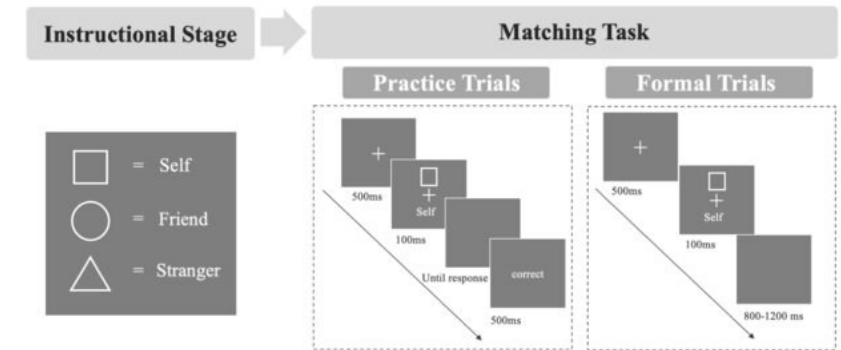
- Self-voices recognition showed:
 - faster RT, higher ACC and enhanced intelligibility (Pinheiro, Lima, et al., 2019),
 - increased attention grabbing at early processing stages (Conde et al., 2015, 2018)
 - higher subjective evaluation of attractiveness (Hughes & Harrison, 2013)
 - higher activations in several brain regions (Allen et al., 2005; Jardri et al., 2007)
 - distinct long-term representations and neural sharpening in temporal regions (Allen et al., 2006; Hosaka et al., 2021).

□ Unresolved Issues

- Self-voice advantages could not be fully separated from familiarity effects (Stolte et al., 2017; Sui et al., 2012)
- Whether external unknown voices elicit analogous advantages (Hughes & Harrison, 2013; Pinheiro et al., 2023)

□ Perceptual Matching Task (PMT)

- Perceptual matching task (PMT) → avoid familiarity confound and extends to novel self-associated stimuli
- Proposed by Sui et al. (2012):
 - learned arbitrary associations between shapes (e.g., circle) and identity (e.g., stranger)
 - then asked to judge whether presented shape-label pairings matched learned ones.
 - Self-associated pairings were processed faster and more accurately, termed “SPE”



□ Auditory SPE

- Voices from unknown speakers elicit bias through mere ownership (Payne et al., 2021; Exp1).
- Similar findings were replicated in Payne et al. (2024) and Kirk et al. (2024, Exp1).
- Self-voice advantages extend to external voices newly associated with self.
- Auditory SPE was not boosted by gender matching (Payne et al., 2021; Exp2)
- Auditory SPE was not influenced by action of self-generation (Payne et al., 2024).



❑ Cross-cultural generalizable or cultural dependent?

➤ Cultural differences in self-construal

- Western cultures → **independent self** and view self as an autonomous entity distinct from others
- East Asian cultures → typically endorse an **interdependent self** and a sense of connectedness
- (J. K. Lee et al., 2023; Sui et al., 2009; Golubickis et al., 2019; Jiang et al., 2019)

➤ Cross-cultural findings of self-advantages

- Cross-cultural generalizability (Golubickis et al., 2019; J. K. Lee et al., 2023)
- **Attenuated or reversed patterns among East Asians** (Jiang et al., 2019; Sparks et al., 2016; Sui et al., 2009)

❑ Research Question

❑ 1. Does self-prioritization in voices present in Chinese participants? → Study 1

- Voice-label matching task, Payne et al. (2021; 2024) and Kirk (2024)

▣ Debates in the Mechanism of SPE

- **Mandatory** (i.e., automatic) product of decisional processing
- Function at **perceptual** level, without conceptual level involved
- Once material (e.g., shapes, mugs) is associated with the self, stimulus prioritization **inevitably** follows (Sui et al., 2013)
- (Humphreys & Sui, 2016; Sui & Humphreys, 2015; Golubickis et al., 2026)

- **Context-dependent and flexible** in its nature
- **Contingent** upon a range of stimulus and task-related factors
- Characteristics of possessions, prior beliefs, etc.
- (Caughey et al., 2021; Golubickis & Macrae, 2023; Jalalian et al., 2024).

Identifying mediators serves as an effective strategy to understand its underlying mechanism (Schäfer et al., 2023).

□ Valence and SPE

□ Tangible Valence-related Properties

- Constable et al. (2021): a greater SPE emerged → associated **happy faces** with self and sad faces with a stranger
- N. A. Lee et al. (2023): self-bias emerged when self was paired with **happy faces** and absent in neutral faces.
- Golubickis et al. (2021): **positive posters** exhibited more reliable self-prioritization compared to negative posters
- Hu et al. (2020): prioritization was stronger once being assigned to the **morally positive aspects of self**
- Schäfer et al. (2023): **negative symbols** attenuated self-bias in the first block

□ Implicitly Valence-related Properties

- Sui and Humphreys (2015b): increasing **stimulus size** enhanced the magnitude of self-bias
- Constable et al. (2021): SPE was larger in the **self-light**/stranger-dark combination
- Vicovaro et al. (2022): self-prioritisation was absent when **asymmetric** shapes became self-associated
- Keil et al. (2023): More pronounced self-bias when **larger numbers** (greater than 5) became self-associated
- Dalmaso et al. (2025): Food stimuli from one's **own culture** elicited a stronger self-prioritisation than foreign foods

□ Valence and SPE

□ Theoretical accounts

- Account A: **integrative theory of self** (Sui, 2016; Sui & Humphreys, 2015a)
- Self functions as an “**integrative glue**” and **binds stimuli to self-concept** (Desebrock, Spence, et al., 2022; Hu et al., 2020). Therefore, information compatibility with the self, especially the alignment with existing self-views, influences SPE (Constable et al., 2021; García-Arch et al., 2024).
- Account B: **shared mechanism** between valence-processing and self-processing (Vicovaro et al., 2022)
- SPE could be partially explained by an **intrinsic positive evaluation** of self-related stimuli (Ma & Han, 2010). Self-associated stimuli quickly **elicit positive affect** (Orellana-Corrales et al., 2023) and get dampened under negative affective states (Feldborg et al., 2021; Sui et al., 2016).



□ Valence and SPE

□ Opposing Evidence

- Account C: **Paralleled processing mechanism** (Liu et al., 2023)
- SPE was unaffected by valence of emotional faces within shapes or induced moods (McIvor et al., 2021; Qian et al., 2020)
- Established self-relevance did not influence facial emotion recognition (Liu et al., 2023)
- No cross-task correlation emerged between self- and emotion-bias (Stolte et al., 2017)

□ Modality Constraints

- **primarily focused on visual modality** and manipulated visual properties (Constable et al., 2021; Golubickis et al., 2021)
- the **salience of prosody in daily communication** (Ding & Zhang, 2023; Lin et al., 2020), effect may extend to the auditory domain.
- higher P2 and LPPs emerging for positive and negative self-uttered words respectively (Pinheiro et al., 2016; Pinheiro et al., 2023)

□ Research Question

□ 2. Does SPE of voices get enhanced by positive prosodic valence? → Study 2

- Valence-manipulated Matching task, Golubickis et al.'s (2021) and Hu et al (2020)

□ Explicit self-esteem and (positive) self-processing

□ Definition

- Explicit self-esteem, defined as an individual's conscious, subjective evaluation of self-worth (Greenwald et al., 2002)

□ Explicit self-esteem and self-processing

- higher explicit self-esteem is associated with **greater attention to self-faces and reduced attention to non-self adjectives**, suggesting an influence of self-esteem on attention allocated to self (Nowicka et al., 2018; Salehinejad et al., 2020).
- explicit self-esteem was **negatively correlated with P2 latency** in negative self-relevant words (Yang et al., 2014).
- explicit self-esteem showed **no significant relationship to self-bias** in matching tasks (Schäfer & Frings, 2019)

□ Multifaceted self-representation

- The minimal and the narrative self Gallagher (2000); (Schäfer & Frings, 2019; Schäfer et al., 2023)
- **Minimal self**: a low-level, pre-reflective self grounded in embodied experience and sensorimotor processes
- **Narrative self**: a higher-order, temporally extended cognitive self involving personal identity, episodic memory
- Self-bias in matching tasks reflects the minimal self with automatic distinction between self and non-self
- (Amodeo et al., 2021; Desebrock, Barutchu, et al., 2022; Moseley et al., 2022)

□ Explicit self-esteem and (positive) self-processing

□ Explicit self-esteem and positive-self processing

- the antecedent role of self-esteem in self-positivity bias (Lin et al., 2003).
- People with higher explicit self-esteem **presented more frequent self-enhancing behaviors** and rated themselves as **better than average in traits** (Campbell et al., 2002; Hepper et al., 2011). Explicit self-esteem was positively correlated with participants' **endorsement of positive traits** (Yang et al., 2016; Zhang et al., 2013)
- Researchers observed **a tendency to prioritize positive self in matching tasks**, reflecting a common bias toward positive self (Constable et al., 2021; Golubickis et al., 2021; Hu et al., 2020). Little attention paid to individual variability (Constable et al., 2021).

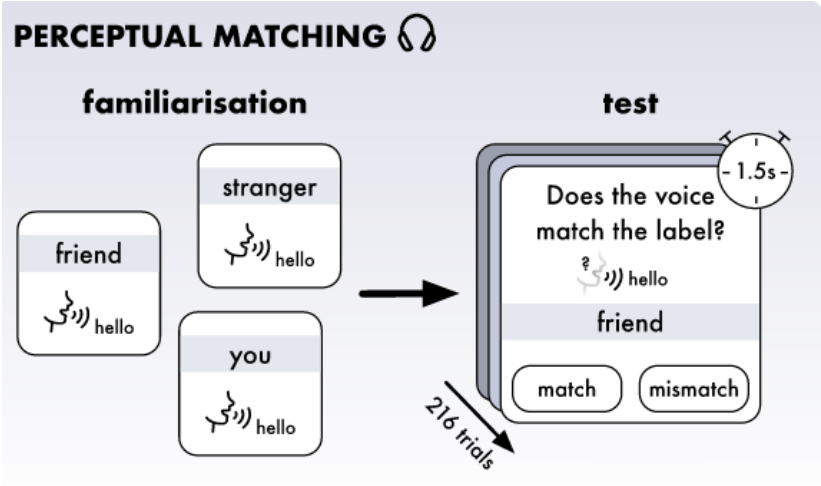
□ Research Question

- Q3: Relationship between with explicit self-esteem and biases for self and positive self in PMT

Research Questions

1. Does self-prioritization in voices present in Chinese participants? → Exp1
- Voice-label matching task, Payne et al. (2021; 2024) and Kirk (2024)
2. Does SPE of voices get influenced by positive prosodic valence? → Exp2
- Modified version of matching task, Golubickis et al.’s (2021) and Hu et al (2020)
3. Relationship between with explicit self-esteem and biases for self and positive self?

Overall Design



	Independent Variable	Trials	Dependent Variable	HDDM	Participants
Exp1	Matching (2) × Identity (3)	$2 \times 3 \times 36 = 216$	RT, ACC, D prime	√	48
Exp2	Valence (2) × Identity (2)	$2 \times 2 \times 2 \times 36 = 288$	RT, ACC, D prime	√	52
Correlation	Rosenburg Self-esteem		Self-bias; Positive self bias	×	48; 52

Study 1: Auditory SPE: Perceptual prioritisation of self-associated voices among Chinese participants



□ Participants and Design

□ Design

- 3 (voice identity: self, friend, stranger) × 2 (trial type: match, mismatch) repeated-measures design

□ Power Analysis

- Based on Kirk & Cunningham (2025), Payne et al. (2021; 2024), a medium effect size of Cohen's $d = 0.50$ was determined for critical comparison between the self and non-self conditions.
- Given $\alpha = 0.05$ and power = 0.80, a requirement of 34 participants to achieve the intended effect size.
- For counterbalanced assignments (gender-congruency), 48 participants (24 males) were recruited.

□ Participants' Recruitment Criteria

- undergraduate or graduate Chinese students at SJTU, with Mandarin as their native language
- born and raised in China, with no residency experience exceeding 4 months in any foreign country
- normal or corrected-to-normal vision and normal hearing
- no history of neurological, psychiatric (mood), or language-related disorders

□ Stimuli and Apparatus

□ Stimuli

- 3 male speakers, 3 female speakers; Mandarin phrase “你好[Hello]”
- drawn from the publicly available AISHELL 2019B-eval dataset (Qin et al., 2020)
- high-fidelity utterances from 86 speakers sampled at 44.1 kHz with 16-bit resolution
- Each speaker recorded the phrase “*Nihao, Mia*” at 3 speaking rates (normal, fast, and slow) in a quiet environment, with 20 utterances per condition. Normal-speed utterances were selected.

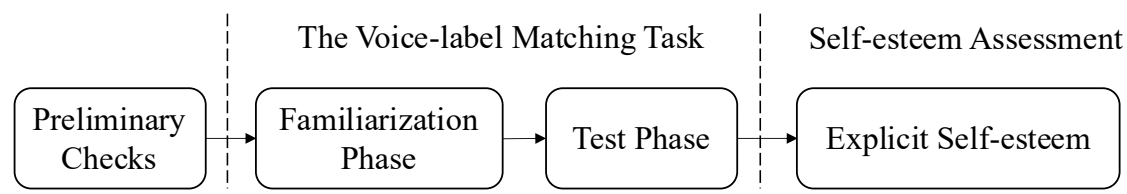
□ Validation Procedure

- 1. an automatic quality assessment using the WV-MOS toolkit → 8 male speakers; 8 female speakers
- 2. extract “*Nihao*” segments, normalize to 70 dB SPL → select the two best-performing exemplars
- 3. 40 participants (20 females) rated attractiveness and trustworthiness on a 7-point Likert scale
- 4. 3 male (M47, M79, M81) and 3 female voices (F25, F34, F70) matched in ratings

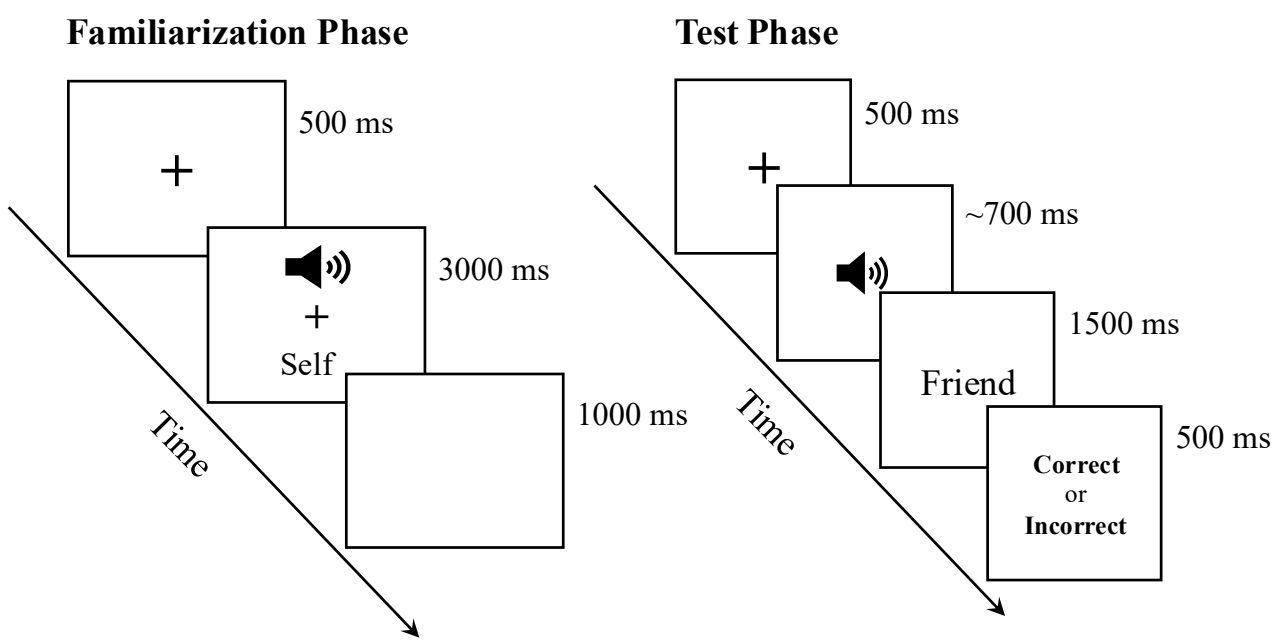


Experiment Procedure

A. Flowchart of the experiment protocol



B. Procedure of the voice-label matching task



	Variables	Trials
Familiarization Phase	Identity (3) × Exemplar (2) × Repetition (2)	3 × 2 × 2 = 12
Test Phase	Practice (12) + Identity (3) × Trial type (2) × Repetition (36)	3 × 2 × 36 = 216

	Description
Familiarization Phase	500 ms fixation cross → 700 ms binaural presentation of voices → a simultaneous display of identity label → label remained on the screen for 3000 ms
Test Phase	500 ms fixation cross → 700 ms binaural presentation of voices → display of identity label for 1500 ms → quick response with key pressing → visual feedback

Match and mismatch trials were in equal numbers and presented in a fully randomized order. Mismatch trials included all types of unmatched voice-label combinations in balanced proportions. Overall accuracy was provided at the end of each block.

□ Data Analysis

□ Measures and Exclusion Criteria

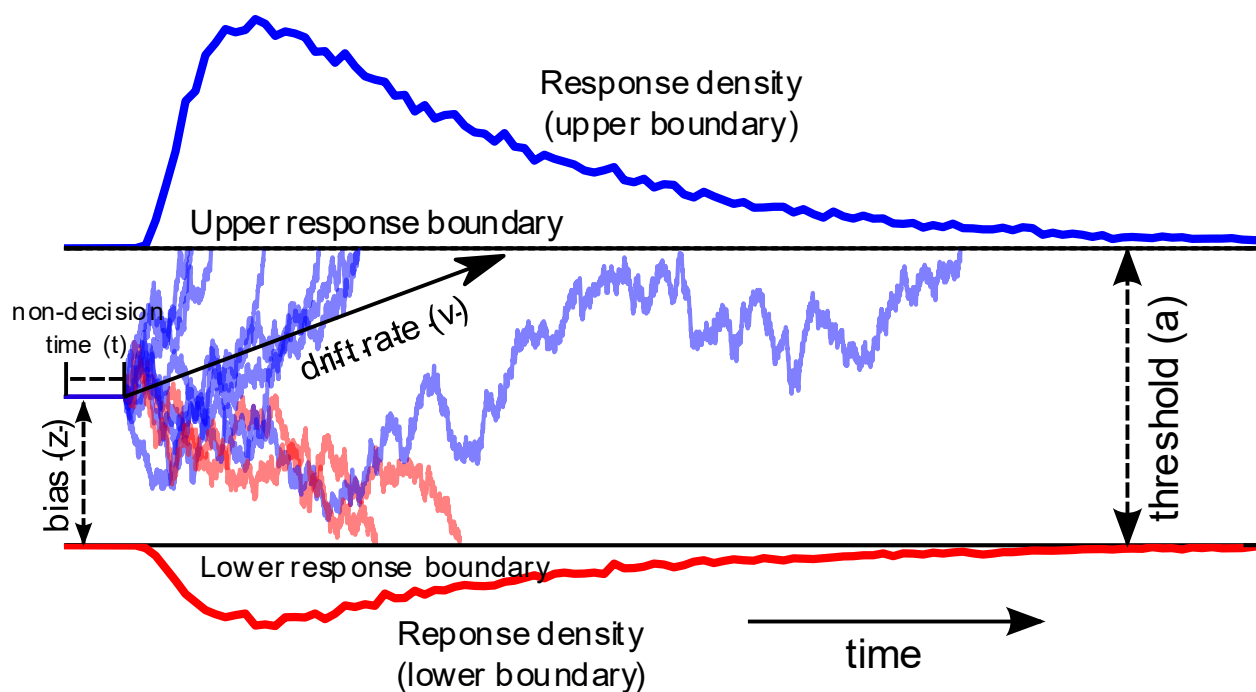
- Reaction Time (RT), Accuracy (ACC), Sensitivity (d')
- RT: the latency between the label onset and the response on each trial
- d' : recoded response as hits or false alarms and corrected hit and false alarm rate
- Trial-level exclusion: shorter than 200ms or longer than 1,500ms, as well as incorrect trials
- Subject-level exclusion: more than 50% of quick responses or ACC below 50% + 95% group CI

□ Statistical Analysis

- RT and sensitivity were analyzed with linear mixed-effects models (LMMs).
- Accuracy was analyzed with binomial generalized linear mixed-effects models (GLMMs).
- The **maximal random-effects structure** justified by the design (Barr et al., 2013) capable of converging.
- Significance determined by **likelihood ratio tests (LRT)**. Post-hoc comparisons with Bonferroni correction.
- Effect size: Partial eta-squared η^2_p for fixed effects and Cohen's d for post-hoc comparisons.

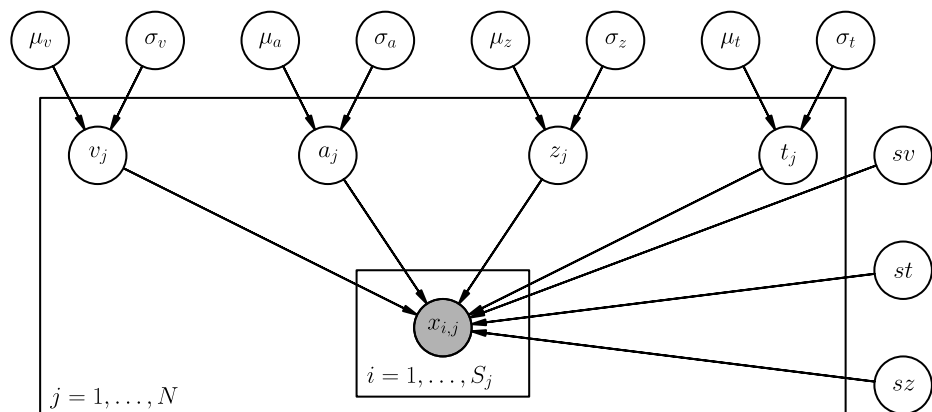
□ Data Analysis

□ Diffusion Modelling



- DDM模型用于描述双选任务中的决策过程。
- 每个选择都由一个上限和一个下限来表示。一个漂移过程随时间累积证据，直到证据累积量跨越其中一个边界并触发相应的反应。
- 模型通常包括下列参数：漂移速率(v)，初始值(z)，非决策时间(t_0)，边界距离阈值(a)。
- 漂移率代表支持或反对特定反应的相对证据积累速度，部分研究认为代表了决策中的刺激偏好。
- 初始值代表决策前的偏向或相对于两个边界的位置，部分研究认为代表了决策中的反应偏好。
- 非决策时间代表决策过程以外的所有时间总和，例如感知、运动启动和执行都需要时间。
- 阈值代表两个边界之间的距离。低阈值反应更快，但噪声对决策影响大，高阈值反应更谨慎。

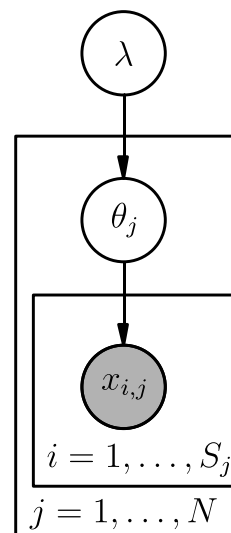
□ Data Analysis



$$\begin{aligned}\mu_a &\sim \mathcal{G}(1.5, 0.75) & a_j &\sim \mathcal{G}(\mu_a, \sigma_a^2) \\ \mu_v &\sim \mathcal{N}(2, 3) & z_j &\sim \text{invlogit}(\mathcal{N}(\mu_z, \sigma_z^2)) \\ \mu_z &\sim \mathcal{N}(0.5, 0.5) & v_j &\sim \mathcal{N}(\mu_v, \sigma_v^2) \\ \mu_{ter} &\sim \mathcal{G}(0.4, 0.2) & ter_j &\sim \mathcal{N}(\mu_{ter}, \sigma_{ter}^2)\end{aligned}$$

$$\begin{aligned}\sigma_a &\sim \mathcal{HN}(0.1) & sv &\sim \mathcal{HN}(2) \\ \sigma_v &\sim \mathcal{HN}(2) & ster &\sim \mathcal{HN}(0.3) \\ \sigma_z &\sim \mathcal{HN}(0.05) & sz &\sim \mathcal{B}(1, 3) \\ \sigma_{ter} &\sim \mathcal{HN}(1)\end{aligned}$$

$$x_{i,j} \sim F(a_i, z_i, v_i, ter_i, sv, ster, sz)$$



$$\mu, \sigma \sim G_0()$$

$$\theta_j \sim \mathcal{N}(\mu, \sigma^2)$$

$$x_{i,j} \sim f(\theta_j)$$

Hierarchical
Bayesian
Estimation

$$f(t | v, a, z) = \frac{\pi}{a^2} \exp\left(-vaz - \frac{v^2 t}{2}\right) \times \sum_{k=1}^{\infty} k \exp\left(-\frac{k^2 \pi^2 t}{2a^2}\right) \sin(k\pi z)$$

$$p(\theta | x) = \frac{p(x | \theta) \times p(\theta)}{p(x)} = \frac{p(x | \theta) \cdot p(\theta)}{\int p(x | \theta) \cdot p(\theta) d\theta}$$

$$p(\theta, \lambda | x) = \frac{p(x | \theta) \times p(\theta | \lambda) \times p(\lambda)}{p(x)}$$

后验分布无法直接计算, 参数空间的高维性, 需估计后验分布形状

Monte Carlo Markov Chain: 以采样方式算出难以直接计算的复杂分布

Monte Carlo: 通过随机采样逼近后验分布 $\lim_{n \rightarrow +\infty} \frac{1}{n} \sum_{i=1}^n x_i = E(X)$

Markov Chain: 构建马尔可夫链, 下个样本仅依赖前个样本, 使样本逐步收敛到目标分布. 去除初始段的不稳定样本, 后半段平稳期的样本分布视为近似服从后验分布

<https://hddm.readthedocs.io/en/latest/methods.html>

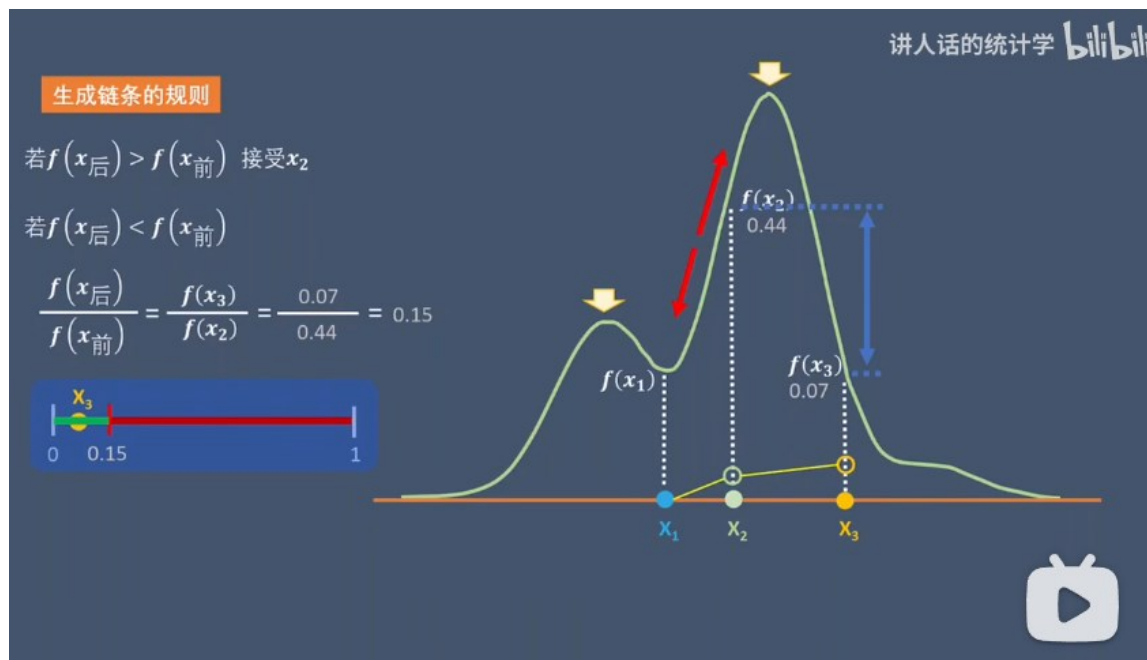
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Study 1: SPE in voices (Data Analysis)

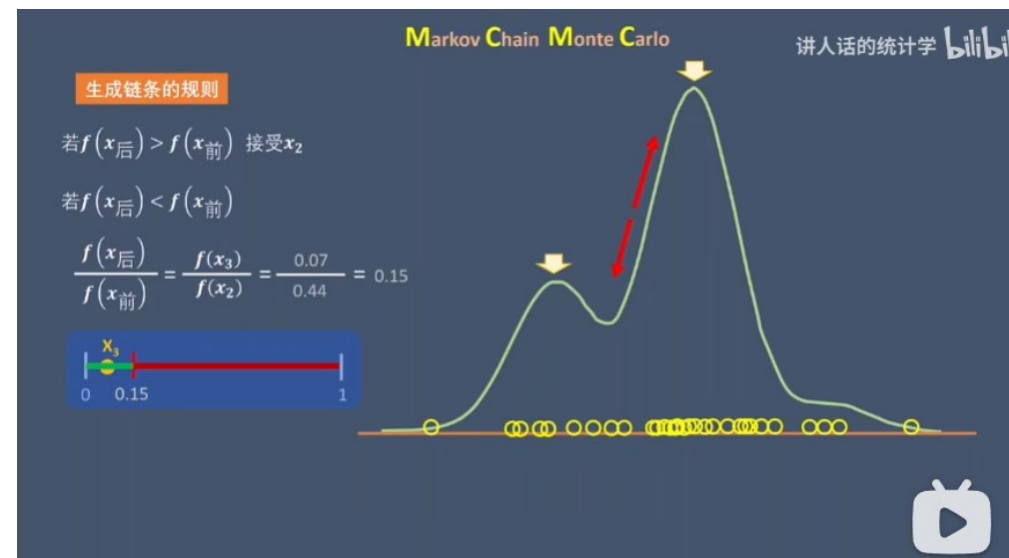
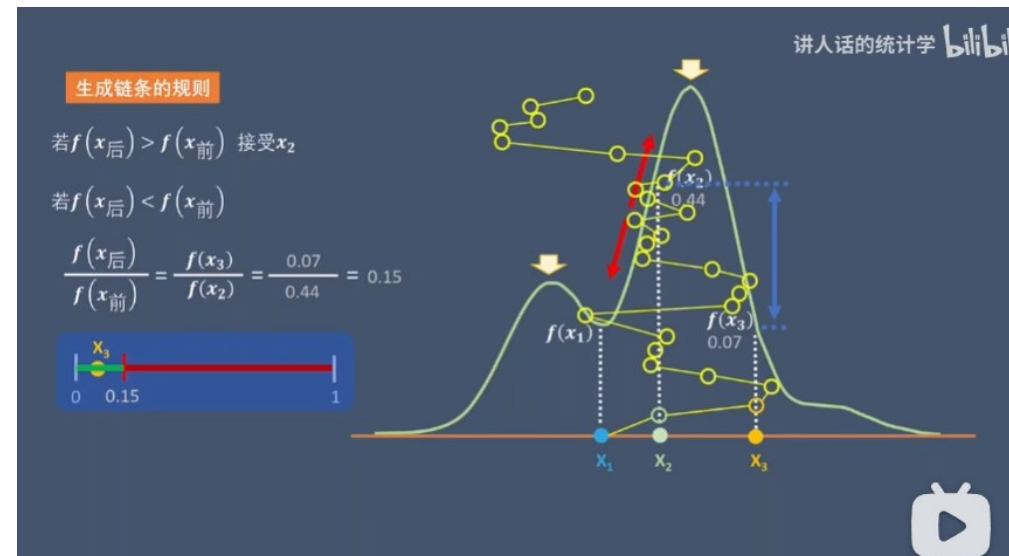


□ Data Analysis



$$f(t | v, a, z) = \frac{\pi}{a^2} \exp\left(-vaz - \frac{v^2 t}{2}\right) \times \sum_{k=1}^{\infty} k \exp\left(-\frac{k^2 \pi^2 t}{2a^2}\right) \sin(k\pi z)$$

https://www.bilibili.com/video/BV1JCCpZYEeH/?spm_id_from=333.1391.0.0&vd_source=38836aa1d9060af599ba1935fcd7efbc



□ Data Analysis

□ Diffusion Modelling

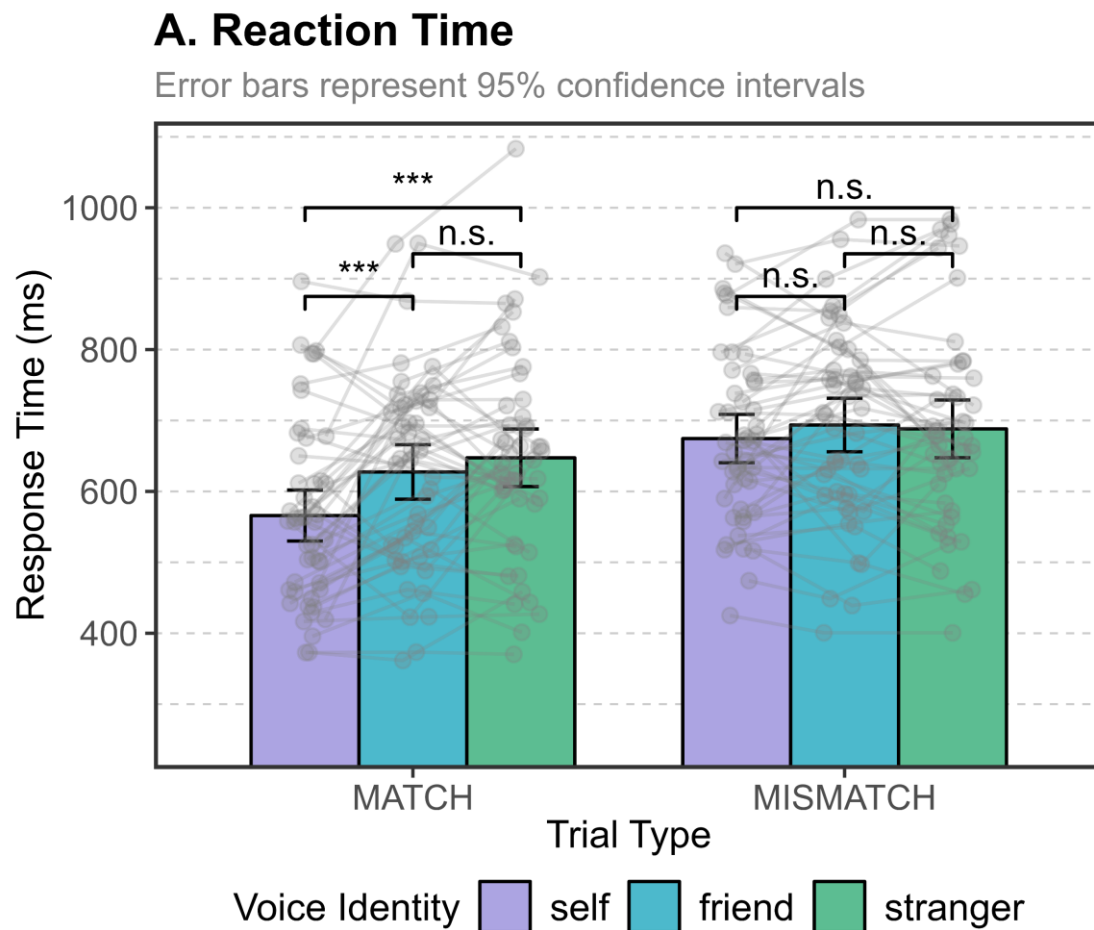
- Drift-diffusion models (DDMs) implemented with Python package HDDM (Wiecki et al., 2013)
- Basic Settings: 1. **response-coded** (the upper threshold referred to “match” response); 2. Bayesian posterior distributions fitted by MCMC with **10,000 samples and 1000 burn-ins**; 3. Convergence assessed via the inspection of **MCMC chains and the Gelman-Rubin (R-hat) statistic** < 1.01 (Pan et al., 2025).
- Model Settings: Conforming to previous studies (Golubickis et al., 2017; Hu et al., 2020)
 - Drift rates (v) varied as a function of voice identity and trial type
 - Nondecision times (t_0) were estimated for match and mismatch trials separately.
 - Starting value (z) varied between two response thresholds, with 0.5 indicating no response bias.
 - Model selection was based on the **deviance information criterion (DIC)**.
 - Six candidate models: 2 one-parameter models; 3 two-parameter models; 1 three-parameter model
 - The model **including three parameters (v , z , and t_0) yielded the lowest DIC value**, indicating the best fit.

□ Correlation Analysis

□ Measures and Exclusion Criteria

- Following Schäfer and Frings (2019), **absolute self-bias scores** were calculated as: $\text{Selfbias} = \text{Self} - \frac{\text{Friend} + \text{Stranger}}{2}$
- Following prior works (Kirk & Cunningham, 2025; Payne et al., 2021), **percentage self-bias scores** were computed as:
$$\text{Self bias} = \frac{\text{Self} - \frac{\text{Friend} + \text{Stranger}}{2}}{\text{Self} + \frac{\text{Friend} + \text{Stranger}}{2}} \times 100$$
- All RT-based scores were multiplied by -1 so that higher values indicated a greater bias.
- Data points exceeding ± 3 SDs from the mean for each measure were excluded.
- One participant was removed from Pearson correlations analysis.

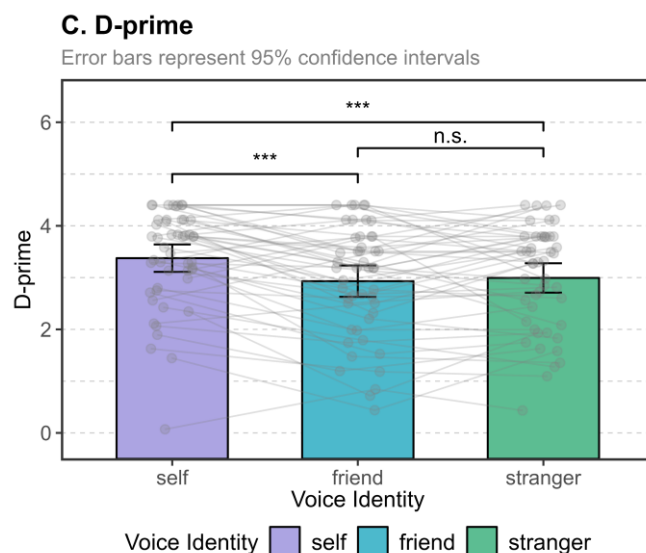
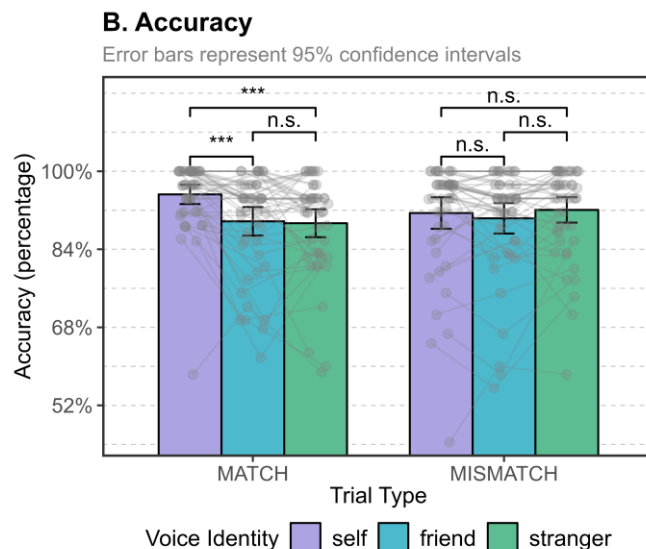
Reaction Time



Effects and Post-hoc

- significant effect of Trial Type [$\chi^2(1) = 73.14$]
 - faster responses in match [$M = 613.10$, $SE = 17.52$] than in mismatch trials [$M = 685.49$, $SE = 17.96$]
- significant effect of Voice Identity [$\chi^2(2) = 15.30$]
 - self-voices [$M = 620.14$] were faster to friend- [$M = 660.25$] and stranger-associated voices [$M = 667.50$]
- significant Voice Identity $\times$ Trial Type interaction
 - Match trials: self-voices < friend-voices and stranger-voices
 - Mismatch trials: no difference among all three identities

Accuracy and d'



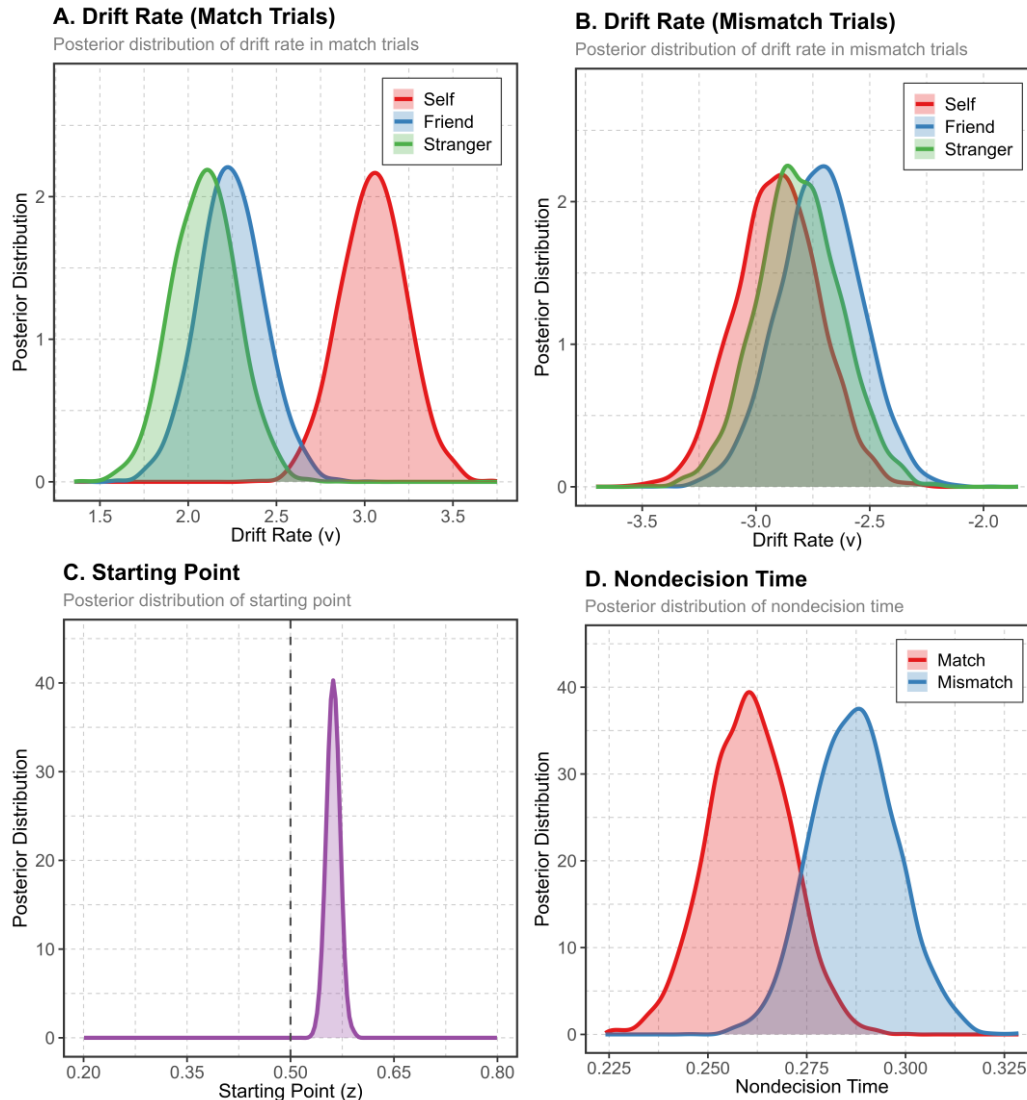
Effects and Post-hoc for ACC

- no significant effect of Trial Type [$p = 0.376$]
- significant effect of Voice Identity [$\chi^2(2) = 32.14$]
 - Accuracy was higher for self-voices than for friend-voices [$\beta = 0.51$, $z = 5.38$, $p < .001$] and stranger-voices [$\beta = 0.42$, $p < .001$, $d = 0.23$]
- significant Voice Identity $\times$ Trial Type interaction
 - Match trials: self-voices $>$ friend-voices, stranger-voices
 - Mismatch trials: no difference among all three identities

Effects and Post-hoc for d'

- significant effect of Voice Identity [$\chi^2(2) = 20.80$]
 - self-voices [$M = 3.38$, $SE = 0.14$] was higher than for friend-voices [$M = 2.93$, $SE = 0.14$] and stranger-voices [$M = 2.99$, $SE = 0.14$]. No significant difference was found between the voices of friend and stranger.

HDDM Analysis



Hypothesis testing of posterior distributions

➤ drift rate (v)

- match trials: drift rates (v) were higher for self-voices ($M = 3.032$, 95% CI [2.671, 3.391]) than for voices of friends ($M = 2.252$, 95% CI [1.895, 2.616], $p[\text{self} > \text{friend}] > 0.999$) or strangers ($M = 2.108$, 95% CI [1.757, 2.461], $p[\text{self} > \text{stranger}] > 0.999$).
- Mismatch trials: no differences among the three identities

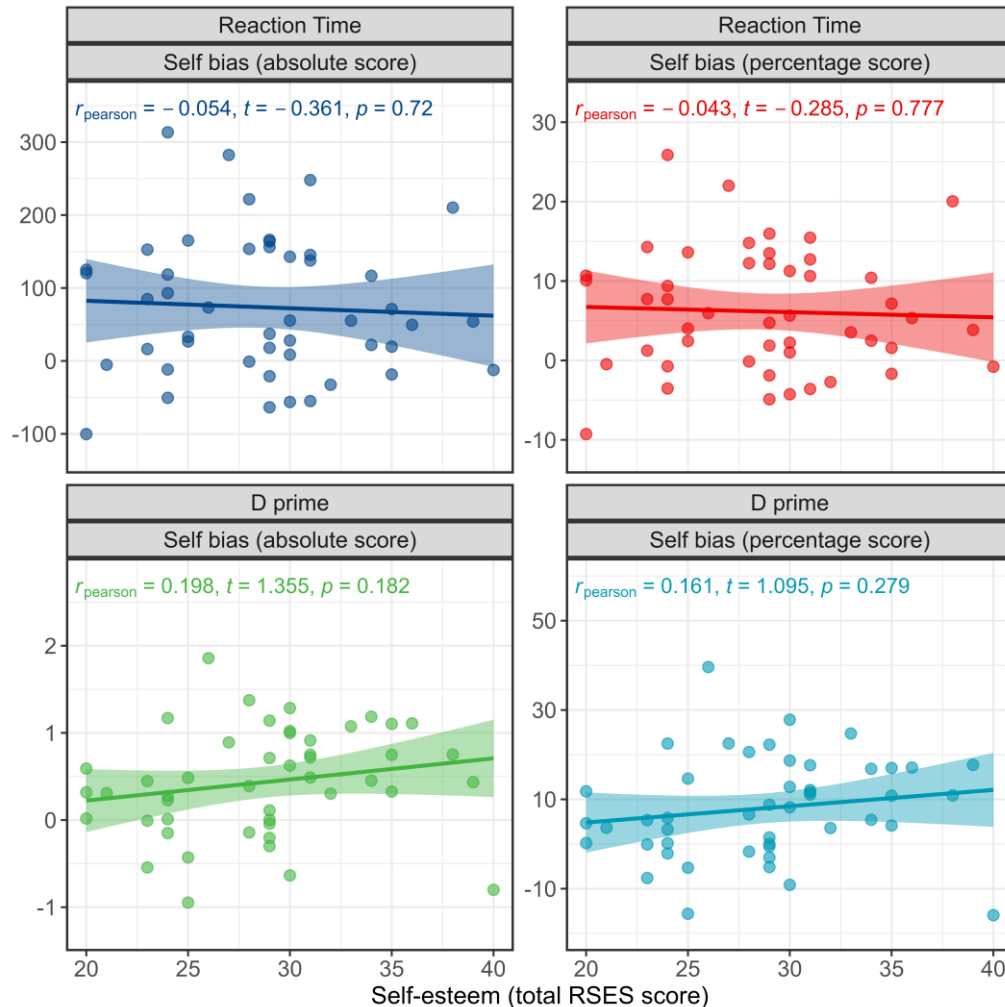
➤ starting point (z)

- a response bias ($M = 0.561$, 95% CI [0.542, 0.581], $p[z > 0.5] = 1.0$)

➤ nondecision times (t_0)

- shorter non-decisional processes in match trials ($M = 0.261$, 95% CI [0.241, 0.281]) than mismatch trials ($M = 0.287$, 95% CI [0.267, 0.308], $p[\text{match} < \text{mismatch}] = 0.962$).

Correlation Analysis



Explicit self-esteem and self-bias

- absolute self-bias scores
- neither of absolute self-bias scores in RT or sensitivity was significantly correlated with RSES scores [RT: $r = -0.054, t = -0.36, p = 0.720$; sensitivity: $r = 0.198, t = 1.36, p = 0.182$].
- percentage scores of self-bias
- both percentage scores of self-bias showed no significant correlations with RSES scores [RT: $r = -0.043, t = -0.29, p = 0.777$; sensitivity: $r = 0.161, t = 1.10, p = 0.279$]

□ Discussion

□ Summary of Findings

- A robust auditory SPE was observed among Chinese participants, evidenced by **faster RTs, greater accuracy, and higher sensitivity** to self-voices relative to friend-voices or stranger-voices in match trials.
- Drift diffusion modelling further uncovered **a stimulus bias**, marked by a higher drift rate for self-voices during decision-making.
- **No significant relationships** between explicit self-esteem and the magnitudes of self-bias.

□ Brief Discussion

- More accurate and faster response in matched trials → align with Payne et al. (2021; 2024) and Kirk (2024)
- Extends earlier findings on self-voice advantages (Conde et al., 2015; Pinheiro et al., 2016; Pinheiro et al., 2023)
- SPE in voices is a robust, fundamental mechanism across cultures (Golubickis et al., 2019; J. K. Lee et al., 2023)
- Explicit self-esteem and self-bias in matching task may represent independent components of self-concept.

Study 2: Valence modulated Auditory SPE: Positive prosodic valence enhanced prioritisation of self-associated voices



□ Participants and Design

□ Design

- 2 (prosodic valence: happy, neutral) × 2 (voice identity: self, stranger) repeated-measures design
- Trial Type: modulatory role of valence on the magnitude rather than the emergence of SPE
- direct examination of relevant theories, control the complexity of mapping learning

□ Power Analysis

- Based on Constable et al. (2021) and Golubickis et al. (2021), the minimal sample size calculation was based on an assumed effect size of $\eta^2p = .14$.
- Given $\alpha = 0.05$ and power = 0.80, 52 participants are required. In total, 54 participants were enrolled.
- Two participants excluded for low accuracy (26 males, 26 females; $M = 22.23$, $SD = 2.78$ years).
- Recruitment criteria were identical to Exp1, and no participants had taken part in Exp1.

□ Stimuli and Apparatus

□ Stimuli

- 2 male speakers, 2 female speakers; Mandarin phrase “你好[Hello]” in happy and neutral prosody.
- Six professional radio broadcasters (3 males, 3 females; age range: 25-30). Major in broadcasting.
- A professional voice actress recorded the phrases beforehand as a prosodic reference.
- Reference audio was played to the speaker, who then imitated the emotional tone on their own.
- For each valence, the session was repeated four to six times, with the two best exemplars retained.

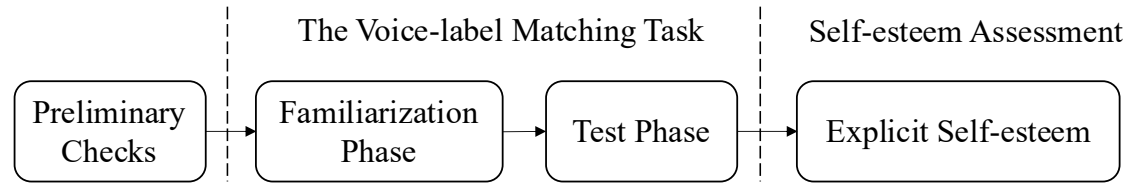
□ Validation Procedure

- 42 participants (22 females) rated **attractiveness and trustworthiness** on a 7-point Likert scale
- Two male (M02, M05) and two female speakers (F03, F06) best matched in ratings.
- 48 participants (23 females) identified **emotional category and rated emotional intensity** on a 7-point Likert scale.
- All stimuli received over 90% accuracy and an average rating of over 4 for emotional intensity.

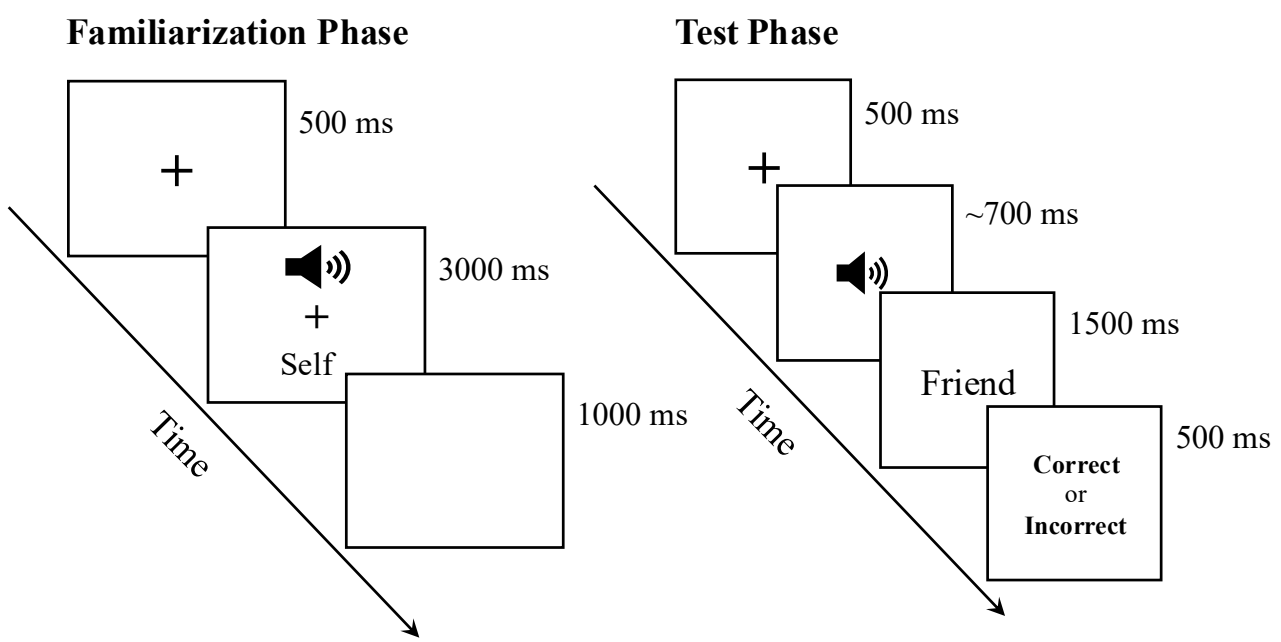


Experiment Procedure

A. Flowchart of the experiment protocol



B. Procedure of the voice-label matching task



	Variables	Trials
Familiarization Phase (2-stage)	Identity (2) × Valence (2) × Exemplar (2) × Repetition (2)	$2 \times 2 \times 2 \times 2 = 16$
Test Phase	Practice (12) + Identity (2) × Valence (2) × Trial type (2) × Repetition (36)	$2 \times 2 \times 2 \times 36 = 288$

	Description
Familiarization Phase	500 ms fixation cross → 700 ms binaural presentation of voices → a simultaneous display of identity label → label remained on the screen for 3000 ms
Test Phase	500 ms fixation cross → 700 ms binaural presentation of voices → display of identity label for 1500 ms → quick response with key pressing → visual feedback

□ Data Analysis

□ Measures and Exclusion Criteria

- Reaction Time (RT), Accuracy (ACC) in **Matched trials**, Sensitivity (d')
- Trial-level: Matched trials shorter than 200ms or longer than 1,500ms, as well as incorrect trials
- Subject-level: more than 50% of quick responses or ACC below 50% + 95% group CI in Matched trials
- Following these criteria, two participants were excluded.

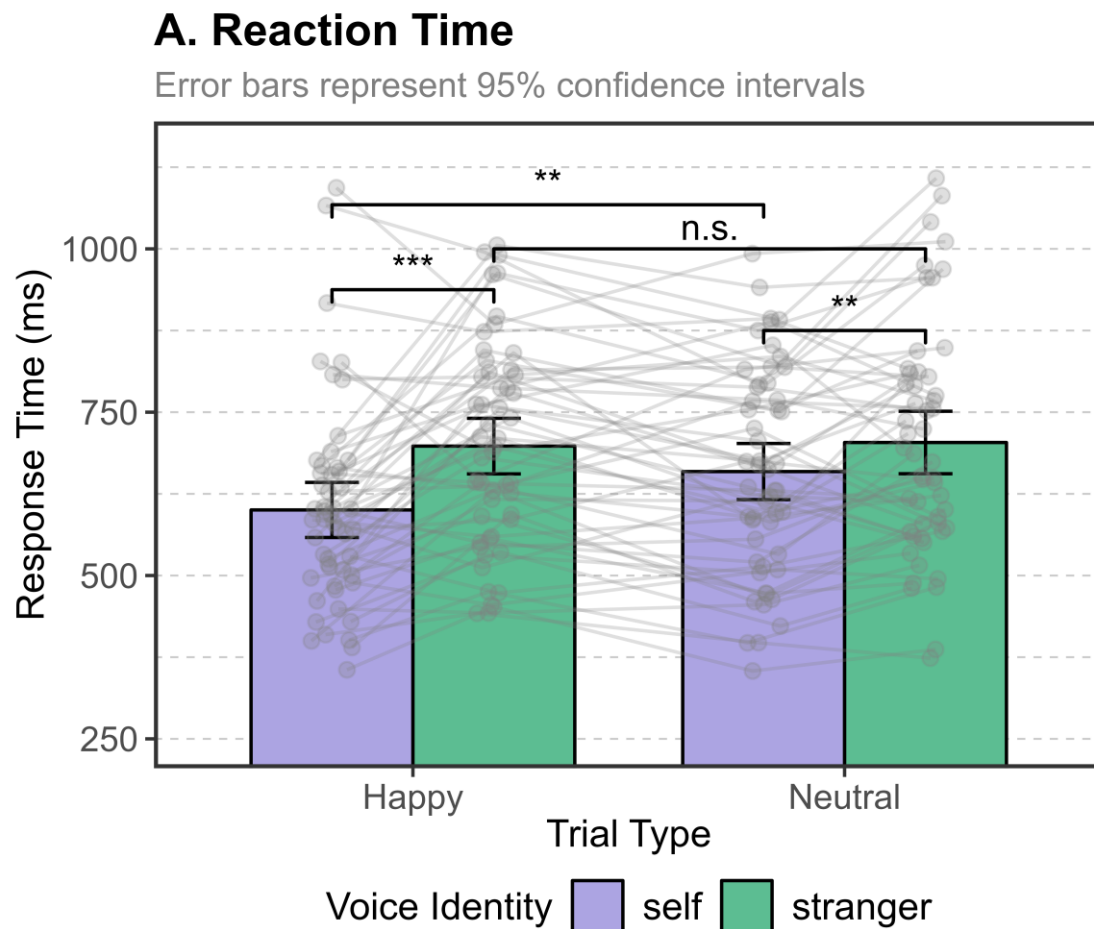
□ Diffusion Modelling

- Model Settings: Conforming to previous studies (Golubickis et al., 2017; Hu et al., 2020)
 - Drift rates (v) varied as a function of voice identity, prosodic valence, and trial type
 - Six candidate models: 2 one-parameter models; 3 two-parameter models; 1 three-parameter model
 - The model including three parameters (v , z , and t_0) yielded the lowest DIC value, indicating the best fit.

□ Correlation Analysis

- **Positive self-bias scores** were derived from the behavioral data under positive valence by substituting variables with “Happy Self” and “Happy Stranger” respectively in the previous formulas.

Reaction Time



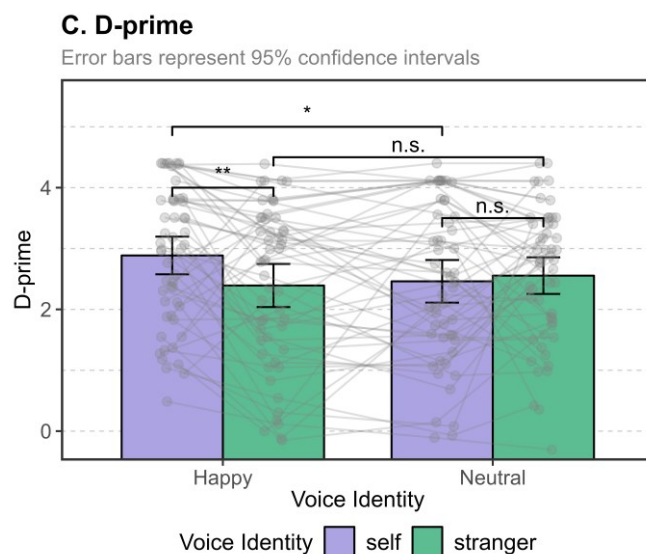
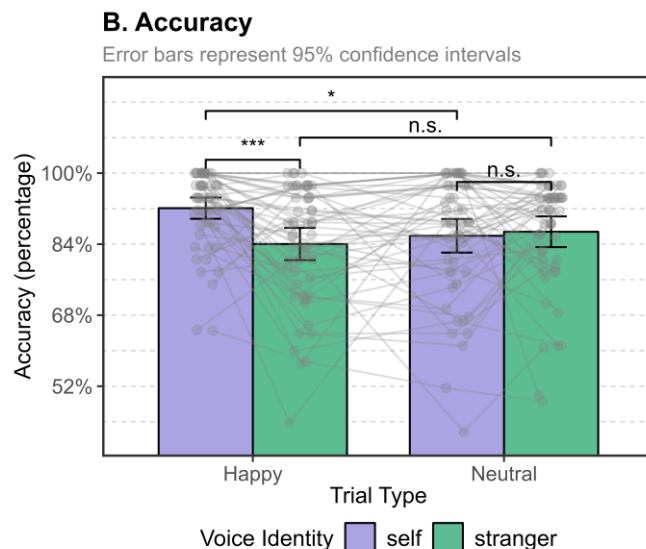
Effects and Post-hoc

- significant effect of Prosodic Valence [$\chi^2(1) = 14.16$]
 - faster responses to happy voices [$M = 648.89$, $SE = 19.32$] than to neutral voices [$M = 679.77$, $SE = 21.88$]
- significant effect of Voice Identity [$\chi^2(2) = 25.40$]
 - Responses were quicker to self-voices [$M = 628.82$, $SE = 20.24$] than to stranger-voices [$M = 699.83$, $SE = 22.13$]
- significant Prosodic Valence $\times$ Voice Identity interaction
 - a larger RT difference between self and stranger for happy voices [$\beta = 97.17$, $SE = 18.28$] than for neutral voices [$\beta = 44.84$, $SE = 13.93$]
 - the effect of prosodic valence emerged for self-voices [$\beta = -57.05$, $SE = 14.71$], with no such effect found for stranger-voices

Study 2: Valence and Auditory SPE (Results)



Accuracy and d'



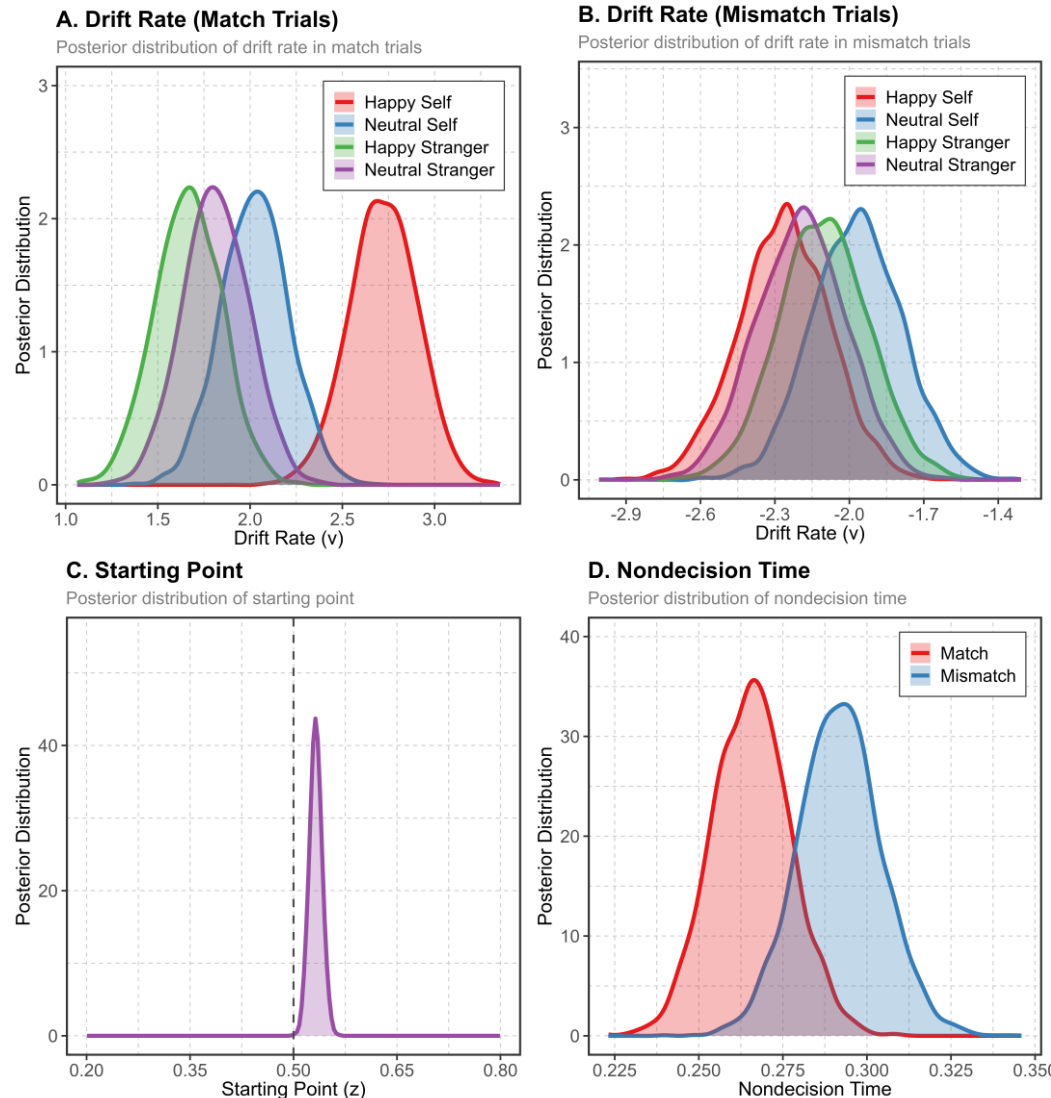
Effects and Post-hoc for ACC

- significant effect of Prosodic Valence [$\chi^2(1) = 13.20$]
 - higher accuracy for self-voices than stranger-voices [$\beta = 0.54$, OR = 1.71, $z = 3.68$, $p < .001$, $d = 0.30$]
- significant Prosodic Valence $\times$ Voice Identity interaction
 - For happy voices, participants responded more accurately to self-voices than to stranger-voices [$\beta = 0.61$]. For neutral voices, the difference between voices of self and stranger was not significant.
 - the effect of prosodic valence was only observed in self-associated voices [$\beta = -0.97$], rather than stranger-associated ones

Effects and Post-hoc for d'

- significant Prosodic Valence $\times$ Voice Identity interaction
 - For happy voices, sensitivity for self was higher than it for stranger [$\beta = -0.49$], whereas no such difference was found for neutral voices. In addition, the valence effect emerged only for self-voices [$\beta = 0.43$]

HDDM Analysis



Hypothesis testing of posterior distributions

➤ drift rate (v)

➤ match trials: drift rate was higher for happy-self ($M = 2.704$) than happy-stranger ($M = 1.682$, $p[\text{happy-self} > \text{happy-stranger}] = 1.0$). No difference was found between neutral-self ($M = 2.032$) and neutral-stranger ($M = 1.827$). Happy prosody facilitated evidence accumulation for self-voices ($p[\text{happy-self} > \text{neutral-self}] = 0.997$), with no such trend emerging for strangers ($p[\text{happy-stranger} > \text{neutral-stranger}] = 0.280$).

➤ Mismatch trials: no differences across conditions [all $p^{\text{Bayes}} < 0.90$]

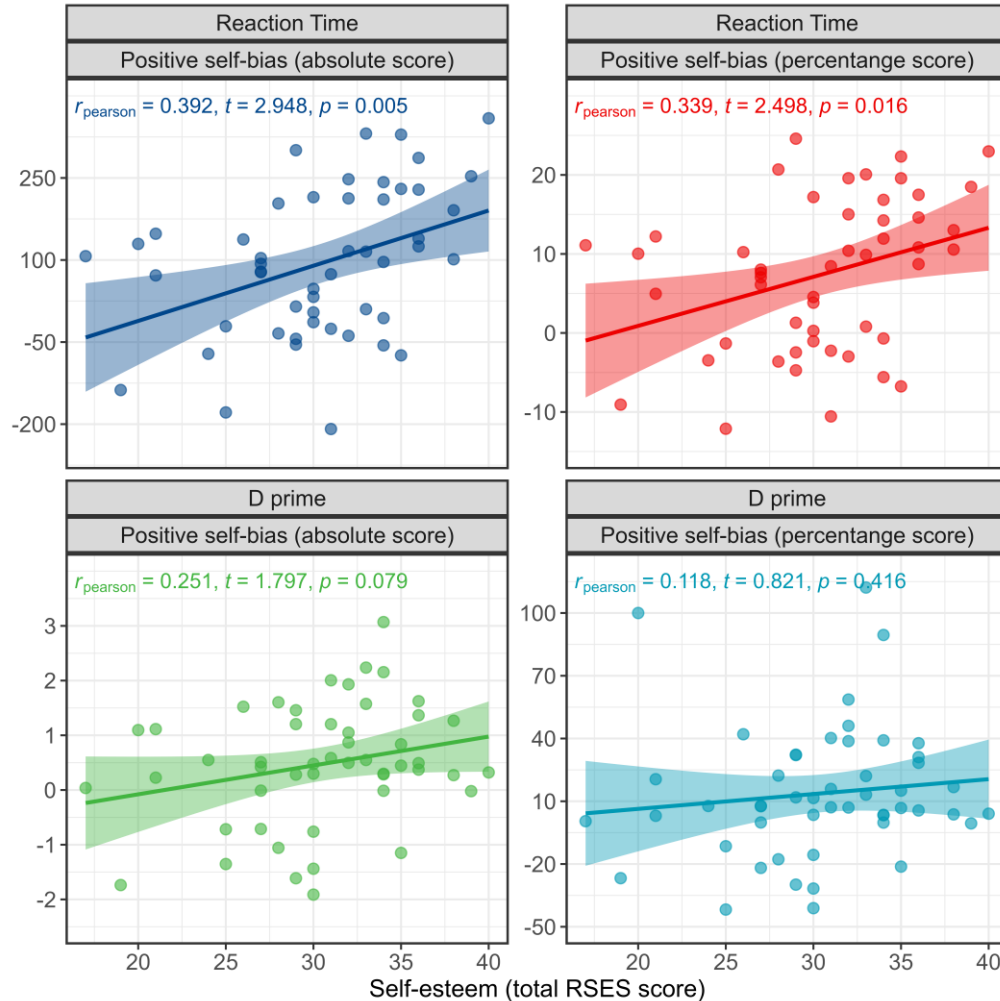
➤ starting point (z)

➤ a response bias ($M = 0.532$, 95% CI [0.514, 0.550], $p[z > 0.5] = 1.0$)

➤ nondecision times (t_0)

➤ marginal evidence for shorter non-decisional processes in match trials ($M = 0.266$, 95% CI [0.245, 0.289]) than mismatch trials ($M = 0.292$, 95% CI [0.268, 0.316], $p^{\text{Bayes}}[\text{match} < \text{mismatch}] = 0.945$).

Correlation Analysis



Explicit self-esteem and positive self-bias

- absolute positive self-bias scores
- Pearson's correlations revealed a significant positive relationship between RSES scores and positive self-bias in RT [$r = 0.392, t = 2.948, p = 0.005$]. The correlation between RSES scores and positive self-bias in sensitivity was marginally significant [$r = 0.251, t = 1.797, p = 0.079$].
- percentage positive self-bias scores
- RSES scores were positively correlated with positive self-bias in RT [$r = 0.339, t = 2.498, p = 0.016$], whereas no significant association was found with positive self-bias in sensitivity [$r = 0.118, t = 0.821, p = 0.416$].

□ Discussion

□ Summary of Findings

- The magnitude of auditory SPE was modulated by the prosodic valence of speech stimuli. Specifically, a **larger SPE was elicited by voices with happy prosody** relative to voices with neutral prosody. Moreover, **the valence effect emerged for self-associated voices selectively.**
- Drift-diffusion modelling revealed that **happy self-voices yielded the fastest evidence accumulation** across all four conditions, indicating facilitated processing for positive self-associated stimuli.
- **Significant positive correlations** between explicit self-esteem and positive self-bias in most measures.

□ Brief Discussion

- evidence for the **flexibility of SPE** → flexible outcomes of decision-making (Golubickis & Macrae, 2023)
- Higher SPE under positive valence → align with Hu et al (2020), etc. Support **integrative self and overlaps**
- Not about denying the parallel-processing account (Liu et al., 2023; Stolte et al., 2017). Could be task constraints.
- Valence effect **only under self condition** → distinctive integrative mechanism of the self
- Significant correlation → the important role in self-positivity bias, addressing **individual differences**

□ Implication

- Individuals with disabilities in their natural voices (e.g., laryngectomy, cerebral palsy, and stroke) rely on voice prostheses (VPs) devices.
- Self-voice misattribution was documented in certain clinical and subclinical populations, such as individuals with schizophrenia or high hallucination proneness (Pinheiro et al., 2019)

□ Limitation

- Exp1: Cross-cultural examination of auditory SPE (eg. Chinese and English Participants)
- Exp1: Self-chosen and randomly-allocated; external and true self-voices. Payne et al. (2021) and Kirk (2024)
- Exp2: Negative condition and self-protection mechanism
- Implicit indices of self-esteem; More measures of traits (social mindfulness, empathy, personal distance)



THANK YOU

